Julio Wall

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EDUCATION

University of Central Florida (UCF) GPA: 3.3
Master of Science in Mechanical/Aerospace Engineering *Graduation: 2027*
Bachelor of Science in Mechanical Engineering *Oct 2021 – Dec 2024*

WORK & LEADERSHIP EXPERIENCE

Pinch A Penny Naples, FL
Pool & Spa Technician / Sales Associate *Oct 2024 – Present*

- Produced high-quality architectural renderings for pool and spa projects, translating design concepts into compelling visual deliverables for client presentations.
- Developed detailed 2D drawings and technical documentation in direct collaboration with the engineering team, ensuring design intent was accurately communicated across all project phases.
- Worked at the intersection of design and engineering, coordinating on spatial layouts, material specifications, and construction details.

University of Central Florida (UCF) Orlando, FL
Undergraduate Teaching Assistant (ULA) *Aug 2024 – Dec 2024*

- Assisted in teaching Intro to Vibrations and Controls under Dr. Sudeshana Pal, providing academic guidance and fostering student understanding of course material.
- Conducted weekly office hours and recitation sessions to support students and clarify complex topics.

Propulsion & Energy Research Laboratory (PERL), UCF Orlando, FL
Injector Design Collaborator *Jan 2025 – Dec 2025*

- Supported design and testing of aerosol injectors for SR-30 and J-85 engines under the STAR program.
- Characterized injector flow and spray behavior at 5–10 g/s to improve droplet uniformity.
- Assisted in evaluating saltwater injection concepts for SAI and Marine Cloud Brightening applications.

Florida Space Institute (FSI) Orlando, FL
Engineer – Project Manager *Oct 2022 – Jan 2024*

- Led a four-member team in innovating experimental hardware, simulating dust aggregate collisions in microgravity to mimic early formation of small icy bodies in the Solar System.
- Designed and iterated 3D-printed test fixtures and hardware components, validating each design through structured lab testing and performance data review.
- Collected, logged, and analyzed experimental data from bi-monthly microgravity regolith simulation runs, identifying trends in particle behavior and collision dynamics.

Engineer – Mechanical Lead

- Led the design and fabrication of a 3U payload for microgravity asteroid soil research.
- Oversaw payload assembly and collaborated with the rocket team to ensure successful integration.
- Conducted testing on a 3U CubeSat, which sustained 90% structural integrity following a 300-foot crash.

Extracurricular

Society of Automotive Engineers (SAE) Aero Design Orlando, FL / Los Angeles, CA
Senior Design – Technical Design and Manufacturing Lead *Aug 2023 – May 2024*

- Designed and manufactured a competition micro-class RC aircraft, applying aerodynamic principles and flight mechanics to develop wings, fuselage, landing gear, payload bay, and control surfaces.
- Presented an innovative micro-class RC airplane design to Lockheed Martin sponsors, representing UCF on an international level at the prestigious 2024 SAE Aero Design Competition.
- Applied principles of flight mechanics and aerodynamics to design and manufacture a micro-class RC airplane, managing the development of wings, fuselage, landing gear, payload, and control surfaces.

Emergency Insights Orlando, FL

Founder & Lead – Entrepreneurship *Jan 2024 – Jan 2025*

- Founded a hazard risk assessment startup using satellite data to deliver property-specific disaster risk reports for homeowners and municipalities.
- Led a cross-functional team, translating complex data into actionable plans and engaging key stakeholders.
- Achieved 4th place out of 60 teams in a high-stakes competition, earning a $2,000 award.

SKILLS, ACTIVITIES & INTERESTS

Languages: English, Spanish

Technical Skills: Python, C, JavaScript, LaTeX

Certifications & Training: Prospective FE License, SolidWorks, Cura, MATLAB, Autodesk Fusion 360

Activities: SHPE, SAE Aero Design Competition, UCF Teaching Assistant, Volunteer Work, Remodeling Projects

Interests: Construction management, Data Compilation, sustainable design, large-scale project coordination, aerospace propulsion, Mechanical Engineering